Education and Information Technologies (2023) 28:13027–13052
https://doi.org/10.1007/s10639-023-11697-6

Engaging EFL students' critical thinking tendency and in-depth reflection in technology-based writing contexts: A peer assessment-incorporated automatic evaluation approach

Chen-Chen Liu[1] · Shi-Jie Liu[1] · Gwo-Jen Hwang[2,3] · Yun-Fang Tu[4] · Youmei Wang[1] · Naini Wang[5]

Received: 4 January 2023 / Accepted: 23 February 2023 / Published online: 22 March 2023

Abstract
With the rapid development of Artificial Intelligence, automatic writing evaluation (AWE) has received much attention from English Foreign Language (EFL) writing teachers. However, the obstacles and potential problems of integrating AWE in EFL writing instruction have yet to be explored. Scholars have indicated that the effectiveness of AWE in EFL writing instruction depends on the learners' depth of reflection. Hence, this study proposes a learning approach that integrates AWE and peer assessment (PA) based on the knowledge-building theory, with the expectation that learners will be able to strengthen their reflections on AWE feedback through PA, and thereby improve their EFL writing performance. To examine the effectiveness of the proposed approach, a quasi-experiment was conducted in a university EFL writing class. One of the classes (33 students) was the experimental group using the PA-AWE approach, and the other class (31 students) was a control group that studied using the conventional AWE approach (C-AWE approach). Findings revealed that the PA-AWE group outperformed the C-AWE group regarding EFL writing performance, learning motivation, critical thinking, and reduced EFL writing anxiety. In addition, a thematic inductive qualitative analysis of the interview data indicated each approach's benefits and learning conceptions.

Keywords Automatic written feedback · Peer assessment · Language learning · Artificial intelligence · English as foreign language

✉ Gwo-Jen Hwang
gjhwang.academic@gmail.com

Extended author information available on the last page of the article

1 Introduction

English is an international language widely used in education and communication (Liu et al., 2021). Writing in English is challenging for most learners due to their poor English expressive skills, lack of vocabulary, and weak grammatical application, making it difficult to accurately translate their thoughts into written expression during the writing process (Challob, 2021; Shang, 2022). Additionally, due to the enormous class size and the heavy pressure on the teacher to review, only some students in traditional English Foreign Language (EFL) writing instruction receive practical guidance and feedback from the teacher in time (Ferris, 2010). Moreover, due to the short classroom time, teachers tend to spend all their time explaining writing skills, making it difficult for students to receive timely correction and guidance on word usage, sentence expressions, and the correct use of grammar in class, thus failing to ensure that their writing skills can improve.

Considering the importance and effectiveness of immediate feedback and the difficulty for teachers to provide immediate feedback in and out of the classroom, researchers and educators have continued to experiment with technology tools to enable real-time feedback (Calvo & Ellis, 2010). As technology has evolved, traditional face-to-face feedback has been replaced by computer-assisted electronic feedback, which in the context of EFL writing instruction refers primarily to automated feedback provided by computers (Pham, 2020). An example is the automatic written corrective feedback (AWCF) provided by automated writing evaluation (AWE) systems. It provides more accurate and flexible automated written feedback through complex natural language processing (NPL) techniques and latent semantic analysis (Bai & Hu, 2017; Li et al., 2015; Warschauer & Ware, 2006). Moreover, by providing adequate and timely feedback and revision advice for students' English writing on a large scale and in a short time, AWE has partially helped teachers to relieve the pressure of correcting (Ranalli, 2018; Zhai & Ma, 2021), which makes it favored by EFL writing teachers.

As a result, it has been introduced into the EFL writing classroom by many researchers, and there have been positive findings (Bitchener, 2012; Ferris, 2010). For example, some scholars have suggested that AWE improved the quality and accuracy of students' writing (Li et al., 2015) by giving them real-time feedback on their errors (Shintani, 2016) and allowing them to immediately correct their language errors in a web-based editing system (Shang, 2022), leading to a greater understanding of the correct form of language to use. At the same time, some studies have argued that AWE feedback gives students a chance for repeated practice and multiple reviews, which can lower learning anxiety and enhance writing skills (Cotos, 2015; Stevenson & Phakiti, 2014). For teachers, AWE can quickly help learners to correct fundamental problems, such as vocabulary and grammar. Thus, teachers can significantly reduce the burden of repetitive correction work and allow students to revise their writing more efficiently in terms of content organization (Ranalli, 2018).

However, researchers have pointed out that AWE still has limitations in EFL writing instruction because it seems to be helpful for revising at the word and sentence level, but not for changing the content and meaning of the writing (Dikli & Bleyle, 2014; Warschauer & Grimes, 2008). The findings of Tuzi (2004) and Diab (2010) also reported that when students refer to the feedback provided by AWE, they tend to focus only on the local level (such as grammar, punctuation, wording, and sentence structure), but not the global level (such as content, organization, tone, and purpose). Research has shown that when students view the content of AWE feedback, they pay more attention to correctness and grading, while tending to ignore the higher-level aspects of content organization and argumentative logic (Fu et al., 2022). Moreover, the revision process often involves simple deletions or superficial error corrections, with little revision of the content of the writing (Guo et al., 2021). And yet, the improvement of English writing skills is not only about learners' proficiency and repeated practice of words, grammar and sentences, but also requires them to clearly express their understanding of controversial topics using logical evidence and facts (Filippou et al., 2021). Previous research has shown that a quality essay requires not only presenting a clear position and following a rational argument, but also responding with evidence to refute the argument, and it emphasises the need for learners to have strong critical thinking (Noroozi et al., 2020).

Therefore, to effectively integrate AWE into EFL writing instruction, it is necessary to adopt appropriate teaching strategies to facilitate learners' deeper reflection on the content of AWE and thus improve the quality of revision of writing content (Liu et al., 2021). Scholars consider peer assessment (PA) as a learning strategy that effectively promotes deep self-reflection and gives learners the opportunity to critically consider alternative arguments and rational reasoning in order to improve the quality of students' writing (Li et al., 2020; Noroozi & Hatami, 2018). It involves rating the performance of peers based on the rubrics provided by the teacher after learning (Topping, 1998). It allows students to reflect on their problems as the person being evaluated, review the rubrics as an evaluator, and give an objective assessment to peers (Tsivitanidou et al., 2011). This will not only help learners understand the teacher's views and grading criteria in depth when commenting on the rubrics to revise their wiring (Chen, 2010), but also provides opportunities for students to observe the performance of peers to facilitate self-reflection (Chang, 2012), and compare the difference between individual expectations and actual performance for further improvement (Cho et al., 2006).

In recent years, Researchers have extensively explored the application of PA in EFL writing instruction in recent years and have obtained positive results. For example, effectively improved grammatical accuracy (Van Beuningen et al., 2012), increased peer engagement (Liou & Peng, 2009), improved learning interest and motivation (Shih, 2011), improved quality of writing content by promoting global revisions (Guardado & Shi, 2007; Yang, 2011), and the development of learners' higher-order thinking, such as the tendency of critical thinking (Wu et al., 2015), have all been found. With the rapid development of smart devices and information and communication technology, technology-assisted PA has gained popularity among researchers because it not only provides a non-threatening learning environment for learners, avoiding the awkwardness of face-to-face PA, but also provides more opportunities for students who are reluctant to participate in assessment due to limited knowledge or cultural background (Jones et al., 2006). Previous research has also shown the advantages of technology-assisted PA over traditional PA. In many ways, technology can scaffold the critical dialogue and argumentation process and guide students to provide a higher-level of quality feedback to their peers in a reciprocal and anonymous manner than traditional PA, which has the potential to maximise the accuracy of peer feedback to fully ensure its effectiveness (Latifi et al., 2021; Latifi & Noroozi, 2021).

Therefore, to address the problem of learners using AWE to focus only on the local-level and to improve the critical thinking of learners of English writing, according to the four stages of EFL writing (idea generation, idea connection, idea advancement, and rise above) based on the theory of knowledge building (Jiang et al., 2022), this study proposed an approach that integrates PA and AWE (PA-AWE approach) into the EFL writing instruction, expecting that learners not only receive immediate feedback and guidance from AWE, but also reflect on the AWE content through the PA learning process, thus improving English writing skills and promoting higher-order thinking skills.

To examine whether the proposed approach can help improve learners' writing performance, learning motivation, and critical thinking and reduce their writing anxiety, the following research questions were asked:

(1) In terms of improving students' writing performance, is the PA-AWE approach more effective than the conventional AWE approach (C-AWE approach)?
(2) In terms of promoting students' learning motivation, is the PA-AWE approach more effective than the C-AWE approach?
(3) In terms of enhancing students' critical thinking, is the PA-AWE approach more effective than the C-AWE approach?
(4) In terms of reducing students' writing anxiety, is the PA-AWE approach more effective than the C-AWE approach?
(5) What are the learning experience and conceptions of the students using the PA-AWE approach and the C-AWE approach, respectively?

2 Literature review

2.1 AWE in EFL writing instruction

With the development of technology, AWE acts as a human scorer in EFL writing instruction because its scoring engine approximates or even outperforms human raters and because of its timely and colossal processing power (Zhai & Ma, 2021). It compares student writing to an extensive informational database by combining statistical modelling and algorithms to extract the writing's linguistic, structural, semantic, and rhetorical features to provide qualitative feedback on grammar, mechanics, style, discourse, and organization (Ranalli et al., 2017). It relieves teachers of the burden of correction and provides learners with more opportunities for independent writing learning, so it is increasingly being used as a learning tool in EFL writing instruction (Liao, 2016).

In recent years, AWE has been widely applied in EFL writing instruction. The study found that AWE not only improved the effectiveness of EFL writing, but also improved learners' perceptions and learning experiences. In terms of enhancing EFL writing performance, research has revealed that AWE can facilitate EFL writing by identifying individual errors and providing feedback on spelling, grammar, sentence, and word usage (Grimes & Warschauer, 2010; Shintani, 2016), enabling students to focus not only on the grammatical aspects of writing, but also on the content and expression of the language (Guo et al., 2021). For the EFL teaching process, incorporating AWE into writing courses dramatically improves classroom efficiency and effectively enhances learners' motivation to write (Zhai & Ma, 2021). Importantly, to improve EFL learners' perceptions and learning experiences of the writing process, it has been found that after receiving timely feedback and clear explanations from AWE, students were able to pay attention to personal errors in their writing (Barrot & Agdeppa, 2021), resulting in significant improvements in grammar, word usage, and spelling, higher accuracy in writing (Wang et al., 2013), and more opportunities for learners to learn independently, even without teacher guidance.

However, many researchers still need to find the effectiveness of applying AWE in learning contexts. First, in terms of the quality of AWE feedback content, its accuracy was satisfactory in terms of linguistic form (e.g., grammar, syntax, collocation, and mechanics) but relatively poor in its discourse-level feedback (e.g., content, cohesion, and structure) (Li et al., 2015). This may cause learners to review their writing after receiving automatic feedback by focusing on superficial features rather than paying enough attention to the proposition's content (Link et al., 2022). Furthermore, the study found that it is difficult to provide students with adequate and meaningful revision suggestions in EFL writing instruction based on AWE alone, because the AWE system typically ignores the social aspects of writing, such as the motivation and emotional factors of learners in the writing and revision process (Zhang & Hyland, 2018). It has also been indicated that incorporating appropriate teaching strategies in the implementation of AWE could help learners improve the above-mentioned problems and help learners improve their reflections on the feedback content to improve their writing performance (Liu et al., 2021).

2.2 PA in EFL writing instruction

PA is a learning strategy that allows learners to evaluate peer performance by grading or providing comments based on rubrics provided by the instructor, which not only strengthens students' problem-solving skills and promotes student-centred learning, but also encourages active learning and deeper understanding (Lin et al., 2021). Learners act in two roles in PA activities: assessor and assessee. For the role of assessor, students are required to rate or comment on the work of peers based on the rubrics given by the instructor. In contrast, for the part of the assessee, students can decide whether or not to accept peers' perspectives and revise their work accordingly (Li et al., 2010). In addition, the process of PA enables students to develop a deeper understanding of the teacher's perspective and rubrics. When students learn in this way, they can reflect on peers' work and reconsider their experience of what they are learning (Andrade & Du, 2005).

In recent years, PA has been widely explored in the field of education, such as natural science courses (Chang et al., 2020), art courses (Lai & Hwang, 2015), medical and nursing courses (Lin et al., 2021), and so on. The findings revealed that PA improved learning achievement and motivation and enhanced learners' reflective skills. Especially in the field of EFL instruction, peer-to-peer feedback and communication not only promoted the learners' oral expression (Cho et al., 2006) and vocabulary use skills (Li et al., 2022), but also improved learners' reading comprehension skills (Jin et al., 2022). It has been indicated that through PA, learners could better reflect on their learning performance and thus achieve better learning outcomes (Hyland & Hyland, 2006).

It is noted that the application of PA in EFL writing instruction has been increasing and showing great potential (Cheng et al., 2015). Applying PA to EFL writing not only exposes students to different writing styles (Shang, 2022) and allows students to help each other solve problems in writing (Shang, 2022), but also offers learners the opportunity to negotiate meaning in writing (Ho & Savignon, 2007), which further helps them develop new ideas and identify linguistic problems (Tuzi, 2004), while in turn promoting learners' motivation and engagement (Shih, 2011). Based on the advantages mentioned above, integrating PA into AWE-supported English writing instruction can help learners to think deeply when using AWE, with the hope of improving their English writing skills.

3 Development of the PA-AWE approach for the EFL writing learning environment

3.1 The learning model for the PA-AWE approach

As a model of knowledge creation metaphor, knowledge building emphasizes continuous idea improvement and the formation of community knowledge (Scardamalia & Bereiter, 1994). Referring to the four stages of idea generation, idea connection, idea advancement, and rising above based on the theory of knowledge building (Jiang et al., 2022), this study constructed a PA-AWE writing learning approach, as shown in Fig. 1. In the four-stage learning process of knowledge building, the instructor first assigns writing tasks through the teacher system, and then students can log onto the learning platform to compose ideas based on the writing topic posted by the instructor and form initial writing ideas; once students have completed and submitted their writing, the learning platform will immediately give feedback on it and randomly assign a peer's essay to each student, who will be required to provide comments and suggestions based on the rubrics given by the instructor. After students have completed the comments, they are required to reflect and revise based on the feedback and suggestions for revision from both the machine and peers, and finally submit their revised writing to the student system.

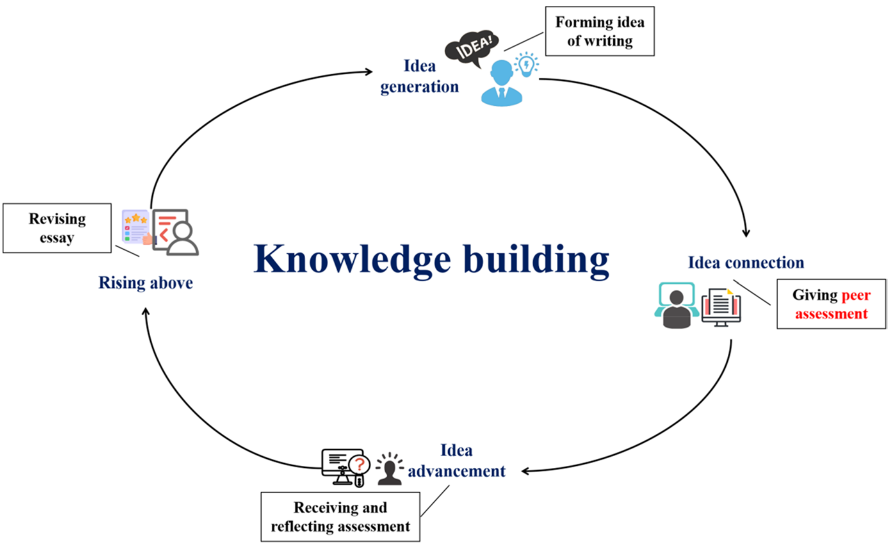

Fig. 1 The learning model for the PA-AWE approach is based on the theory of knowledge building

3.2 The PA-AWE learning environment

Figure 2 shows the structure of the PA-AWE writing learning environment, which consists of a student system, a teacher system, and a system database. In the student system, they can use the learning platform for writing, reflecting, and revising. After viewing the writing assignments posted by the instructor, students could submit their paper to the learning platform and then receive automatic feedback reports from the platform and peer comments to revise their writing. In the

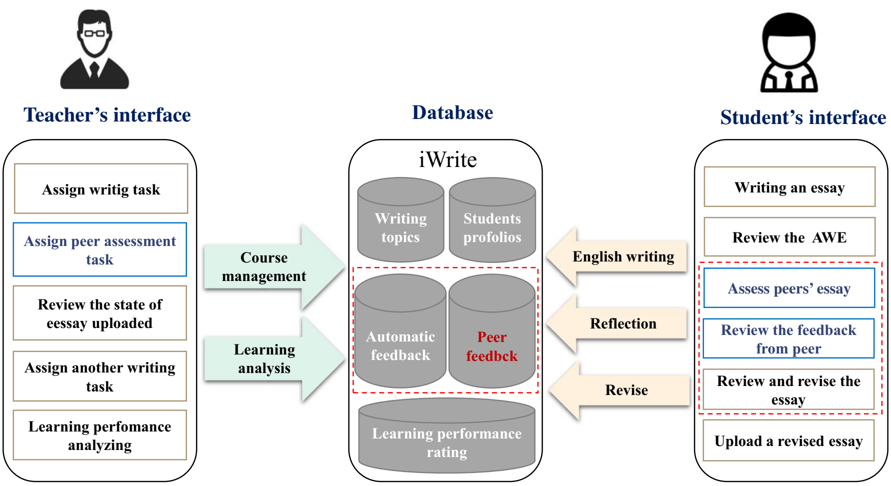

Fig. 2 The system structure of PA-AWE learning environments

teacher system, they can use the learning platform for course management and learning analysis. They can post online writing tasks on the platform and conduct PA during the learning process. Students' revision process could be recorded in the learning platform. In the system database, the learning platform has recorded each student's writing time, writing performance, and revision process. To better comprehend each student's learning situation, the instructor can review each student's essay in the database, and then give timely personalized instruction and analysis of the learning performance.

It has been illustrated in detail how the learning platform functions in Fig. 3. The task management interface allows the instructor to post new writing tasks and view students' writing performance and peer assessments at any time, enabling them to adjust their teaching methods according to students' scores and performance. In the learning performance analysis interface, the learning platform summarizes the average writing score, the highest and lowest scores, and the types of common errors made by students during this writing task, helping the instructor make effective classroom adjustments, knowledge explanations, and personalized instruction based on the problems at any time.

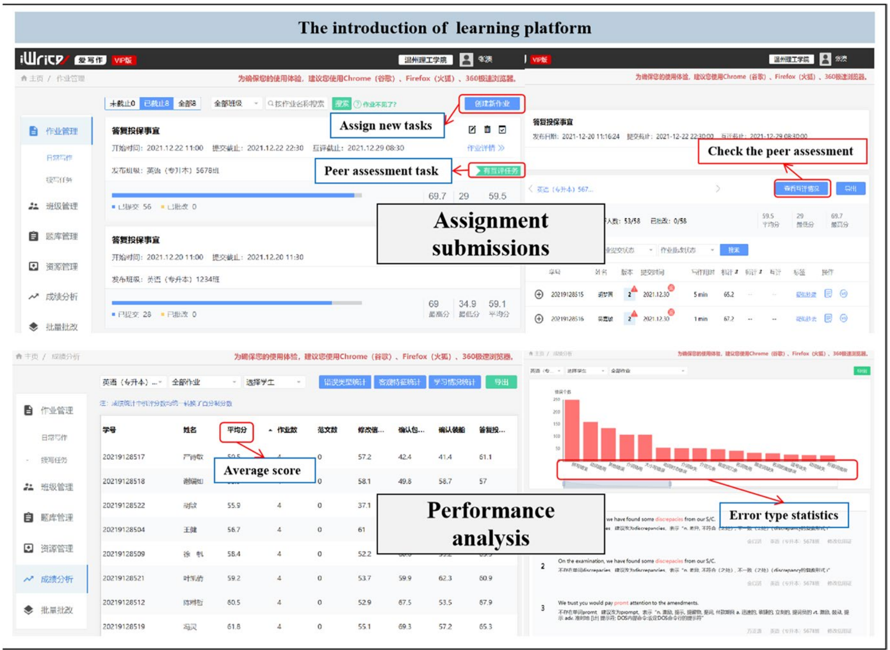

Fig. 3 The introduction of the learning platform

3.3 The learning activities with the PA-AWE approach

3.3.1 The AWE content

Figure 4 illustrates the interface of the AWE report that learners receive on the mobile learning platform, including the total writing score, the scores for each dimension, and the detailed revision suggestions. After receiving the student's writing, the learning platform will grade the student's essay with a total score (up to 100) based on a built-in algorithm. It will also rate the student's writing in four dimensions (out of 5 stars): linguistic, content, article structure, and grammar. When students check their essays, they can see the alternative expressions for words and phrases provided by the platform in terms of language use in the essay, as well as assessments of the relevance and coherence of the content.

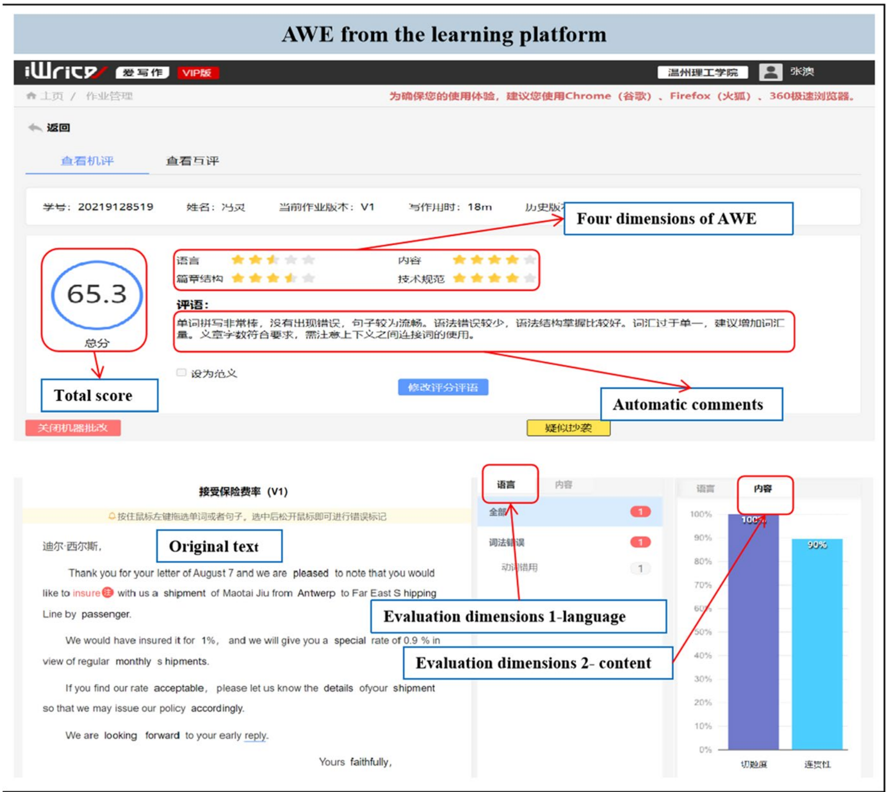

Fig. 4 AWE from the learning platform

3.3.2 The learning process with the PA-AWE approach

In the current study, a technology-supported learning environment is developed that progressively guides learners to think deeply during the writing process and prompts them to reflect on the automated feedback content and rubrics accordingly. Figure 5 illustrates the process of student writing using the PA-AWE approach. At first, students receive writing assignments from the instructor on the learning platform, form writing ideas, then complete the report and submit it. Afterwards, students are randomly assigned to give PA to others and are required to check the rubrics given by the instructor to comment on peers' writing and provide offers. Later, they need to review the content and suggestions for revision passed by the machine and peers to reflect on how to revise and optimize their writing. Finally, they are asked to reconstruct and submit the new report.

4 Method

To explore the effects of the PA-AWE approach on students' EFL writing performance, this study collected both quantitative and qualitative data to analyze the results of the approach on students' learning motivation, critical thinking, and English writing anxiety.

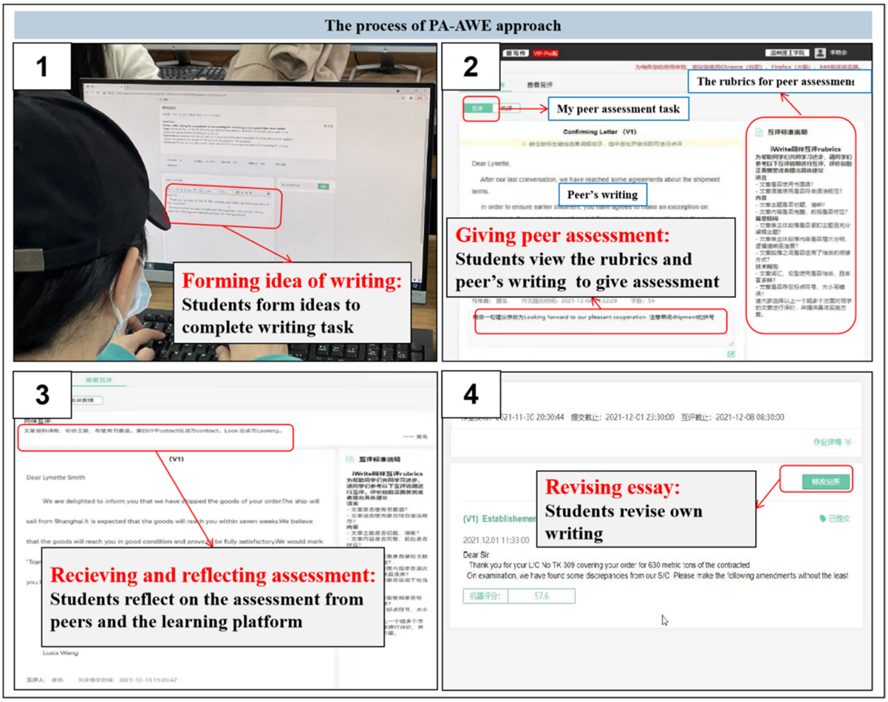

Fig. 5 The process of the PA-AWE approach

4.1 Participants

The participants of this study were 64 students from a university majoring in English, and their average age was 20 years. A total of 33 students in the experimental group used the PA-AWE approach, while 31 students in the control group used the C-AWE approach. The experimental and control groups of the study were taught by the same EFL writing teacher with 5 years of teaching experience. All participants knew the purpose before the experiment, and participated in the activity voluntarily. They knew they could withdraw at any time during the investigation.

4.2 Measuring Instrument

The measurement instruments used in this study included a pre- and post-English writing test, questionnaires on students' learning motivation, critical thinking, and English writing anxiety, and interviews on their learning experiences after the learning activities.

The rubric of EFL writing performance To ensure validity and reliability in measuring participants' EFL writing performance, the writing test items for this study were selected from the CET4 (a Chinese National English Proficiency Test). With reference to Baker et al. (2020), the rubric of writing performance was developed and examined by two experienced teachers; it consists of four dimensions: Argument and support, organization, vocabulary, and language structure. The total score for both pre- and post-tests was 20 points, with each dimension accounting for 5 points each. Two teachers gave scores based on this criterion and the internal consistency was 0.81.

Learning motivation The learning motivation questionnaire was adapted and slightly modified from Wang and Chen (2010) and consists of six items (three for intrinsic motivation and three for extrinsic motivation). For example, one of the items for intrinsic motivation was, "In English writing courses, I prefer challenging content because I can learn new things that way." One of the extrinsic motivation items was, "I want to do well in my English writing course because it is important to show my ability in front of friends, teachers, or other people." The Cronbach's alpha value in the current study was 0.91, indicating that the entire scale reaches acceptable internal consistency reliability (Cohen, 1998).

Critical thinking The critical thinking questionnaire was modified by Lin et al. (2019) and consisted of six items. The questionnaire used a 5-point Likert scale ranging from 5 (strongly agree) to 1 (strongly disagree). For example, one of the items is "When faced with a practical problem, I can assess which opinion is the most reasonable one." The higher the score, the more critical thinking tendencies students have in the EFL writing courses. The Cronbach's alpha value for the current study was 0.96.

Writing anxiety The writing anxiety questionnaire is adapted from Cheng (2017) and consists of nine items. The questionnaire used a 5-point Likert scale ranging from 5 (strongly agree) to 1 (strongly disagree). For example, "When I write in English, I often worry that my writing performance is worse than others." and "When I write in English, I often feel my heart pounding." The Cronbach's alpha value for the current study was 0.94.

Interview outline and coding scheme The interview outline was designed and slightly modified by Hwang et al. (2009). The outline consisted of seven semi-structured interview questions, and a total of 22 students (11 in each group, coded E01-E11 and C01-C11 respectively) were invited to participate in the interviews. The purpose of the interviews was to investigate the strengths and weaknesses of the learning tools and learning strategies from the students' perspectives. The results of the discussions are expected to provide qualitative evidence to explain the findings of this study. The interviews were analyzed by two experienced researchers, and for those classifications that were inconsistent, further review and discussion was carried out until agreement was reached. The interview questions were as follows:

(1) What do you think you have gained most from using this approach? Please give a specific example.
(2) How is the proposed learning approach different from your previous (or expected) English writing course? Do you think it is effective? Why? How has this helped you?
(3) In general, what are the benefits of this approach to learning?
(4) What needs to be improved for this learning approach (e.g., the system functionality or interface design)? Please give us an example.
(5) Would you like to use this way to learn again in the future? For what subjects? Why? Why are these subjects suitable?
(6) Would you recommend this learning approach to other students? Why do you think they need to learn this way? Or would they want to learn this way?
(7) Would you recommend other instructors to use this teaching approach? Why do you think they need this teaching approach? Or would they be willing to use this teaching approach?

4.3 Experiment procedure

Figure 6 demonstrates the experimental procedure for this study. The experiment was conducted in an EFL writing class for 4 weeks, with 90 min of face-to-face instruction per week. During the experiment, the course covered four writing topics, which were developed by experienced instructors. English pre-writing tests, learning motivation, critical thinking, and English writing anxiety pre-test questionnaires were implemented to ensure that all students started with equivalent English writing skills and knowledge. Before the experiment, the instructor introduced the writing system and the four writing topics for learning activities. During the 4 weeks, learners practiced their EFL writing skills on the AWE learning platform, gave

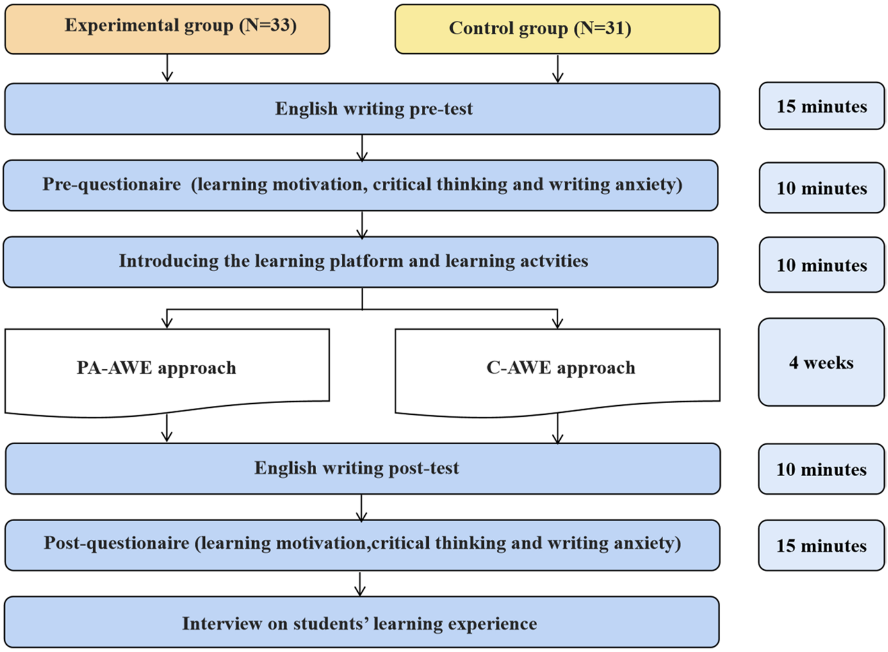

Fig. 6 The experiment procedure

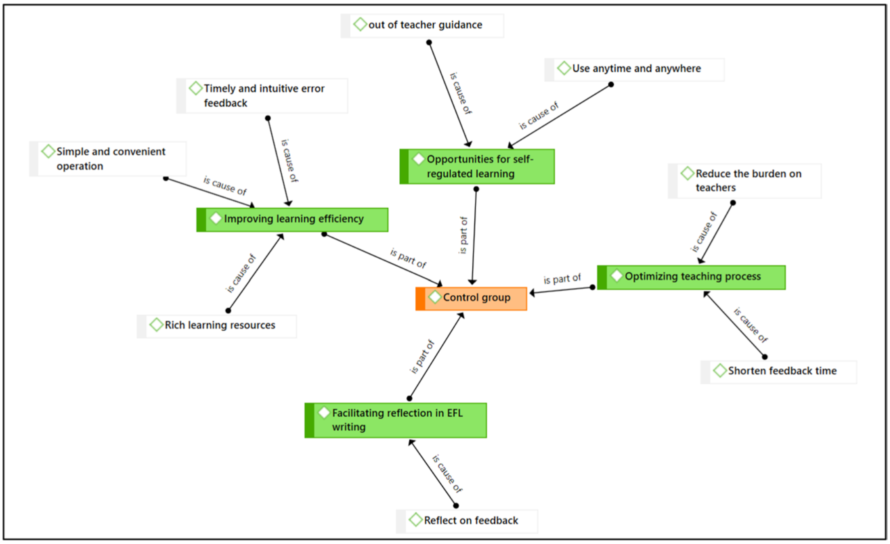

Fig. 7 A conceptual framework for learning gains of students in the control group

assessments for peers, and received assessments from peers. At the end of the learning session, students were asked to take a post-test and complete the post-questionnaire. Furthermore, 22 students from both the experimental and control groups were invited to take part in an interview to investigate their learning experience with the proposed learning method. The students in the experimental group practiced writing using the PA-AWE approach. The instructor would post weekly writing tasks on the learning system in advance and randomly assign learners to give PA for others, and then students would complete the tasks and revise their writing based on AWE feedback and suggestions from peers. Students in the control group used the C-AWE approach for writing practice. Similarly, after completing the writing task, the learners revised their writing based on the AWE feedback, however, without peer comments and suggestions.

5 Results

5.1 Analysis of EFL writing performance

For the EFL writing performance, the results of the Levene test for the quality of variances indicated that the assumption of homogeneity of variances in the groups was satisfactory ($F = 1.164$, $p = 0.258 > 0.05$). Therefore, ANCOVA could be performed for analysis. Table 1 shows that, after excluding the effects of the pre-questionnaire, the PA-AWE group was significantly better than the C-AWE group in EFL writing performance ($F = 7.126$, $p = 0.010 < 0.05$). The effect size for the total score was medium ($\eta^2 = 0.105$) (Cohen, 1988). The standard errors and adjusted means of students' writing performance with the PA-AWE approach were 0.318 and 16.664, while those with the C-AWE approach were 0.307 and 15.458. The results showed that the PA-AWE approach was significant for improving EFL writing performance for students.

5.2 Analysis of learning motivation

For learning motivation, the results of the Levene test for the quality of variances indicated that the assumption of homogeneity of variances in the groups was satisfactory ($F = 1.430$, $p = 0.237 > 0.05$). Therefore, ANCOVA could be performed for analysis. Table 2 shows that, after excluding the effects of the pre-questionnaire, the PA-AWE group was significantly better than the C-AWE group in

Table 1 The ANCOVA results for learning performance

Group	*N*	Mean	*SD*	Adjusted mean	Std. error	*F*	η^2
Experimental	33	16.306	2.817	16.664	0.318	7.126*	0.105
Control	31	15.839	2.549	15.458	0.307		

$^*p < 0.05$

Table 2 The ANCOVA results for learning motivation

Group	N	Mean	SD	Adjusted mean	Std. error	F	η^2
Experimental	33	23.03	3.264	23.093	0.664	7.204**	0.106
Control	31	20.68	5.671	20.610	0.644		

**$p<0.01$

learning motivation ($F=7.204$, $p=0.009<0.01$). The effect size for the total score was medium ($\eta^2=0.106$) (Cohen, 1988). The standard errors and adjusted means of students' learning motivation with the PA-AWE approach were 0.664 and 23.093, while those with the C-AWE approach were 0.644 and 20.610. The results showed that the PA-AWE approach was significant in promoting learning motivation for students.

5.3 Analysis of critical thinking

For the critical thinking tendency, the results of the Levene test for the quality of the variances indicated that the assumption of the homogeneity of variances in the groups was satisfactory ($F=0.301$, $p=0.585>0.05$). Therefore, ANCOVA could be conducted for analysis. Table 3 demonstrates that, after excluding the effects of the pre-questionnaire, the PA-AWE group was significantly better than the C-AWE group in critical thinking ($F=5.532$, $p=0.022<0.05$). The effect size for the total score was medium ($\eta^2=0.083$) (Cohen, 1988). The standard errors and adjusted means of students' critical thinking with the PA-AWE approach were 0.593 and 24.238, while those with the C-AWE approach were 0.612 and 22.230. The results showed that the PA-AWE approach significantly enhanced students' critical thinking.

5.4 Analysis of writing anxiety

For the EFL writing anxiety, the results of the Levene test for the quality of variances indicated that the assumption of homogeneity of variances in the groups was satisfactory ($F=0.047$, $p=0.830>0.05$). Therefore, ANCOVA could be adopted for analysis. Table 4 shows that, after excluding the effects of the pre-questionnaire, the PA-AWE group scored significantly lower than the C-AWE group on writing

Table 3 The ANCOVA results for critical thinking

Group	*N*	Mean	*SD*	Adjusted mean	Std. error	*F*	η^2
Experimental	33	24.06	3.724	24.238	0.593	5.532*	0.083
Control	31	22.42	4.145	22.230	0.612		

*$p<0.05$

Table 4 The ANCOVA results for writing anxiety

Group	*N*	Mean	*SD*	Adjusted mean	Std. error	*F*	η^2
Experimental	33	24.48	4.691	24.536	0.837	5.042*	0.076
Control	31	27.29	6.532	27.236	0.863		

$^*p < 0.05$

anxiety ($F = 5.042$, $p = 0.028 < 0.05$). The effect size for the total score was medium ($\eta^2 = 0.076$) (Cohen, 1988). The standard errors and adjusted means of the students with the PA-AWE approach were 0.837 and 24.536, while those with the C-AWE approach were 0.863 and 27.236. The results showed that the PA-AWE approach was significant for reducing students' writing anxiety.

5.5 Interview results

The interview data for this study involved content analysis based on a transcription of the recorded data and inductive qualitative analysis using ATLAS.ti.22. With reference to Creswell and Poth (2016), open coding was adopted in the current study. We grouped items in the code list with similar meanings into groups to organize new categories, which were then summarized into themes and subthemes for the interview data. During the coding process, 162 codes were identified, of which 101 came from the experimental group and 61 came from the control group. After analysis, five themes and 12 subthemes were concluded for the experimental group, while four themes and eight subthemes were concluded for the control group. In axial coding, links were made between the themes represented by the codes to describe the relationship between different themes and subthemes, which were presented in the form of networks. Figures 6, 7, and 8 show the interview results of the experimental and control groups, respectively. As seen in the two figures, the researchers classified the text data as apply codes (white) and abstract codes (green), which were connected in the code-code relation of "is cause of." Also, five abstract codes were connected in the code-code relation of "is part of" to present the relationships between the abstract codes.

To answer research question 5, five themes were identified after comparing the results of the analysis in Figs. 7 and 8.

5.5.1 Theme-1: Optimizing the teaching process

Integrating AWE into EFL writing instruction optimizes the teaching and learning process and improves classroom efficiency. It not only helped teachers reduce the correction burden (frequency of the experimental group: 5; frequency of the control group: 5), but it also helped students shorten the waiting time for feedback (Frequency of the experimental group: 8; Frequency of the control group: 5). The majority of students said: "Teachers have a heavy teaching load, and it would be difficult to give individual feedback on errors if they were revising one by one; using the

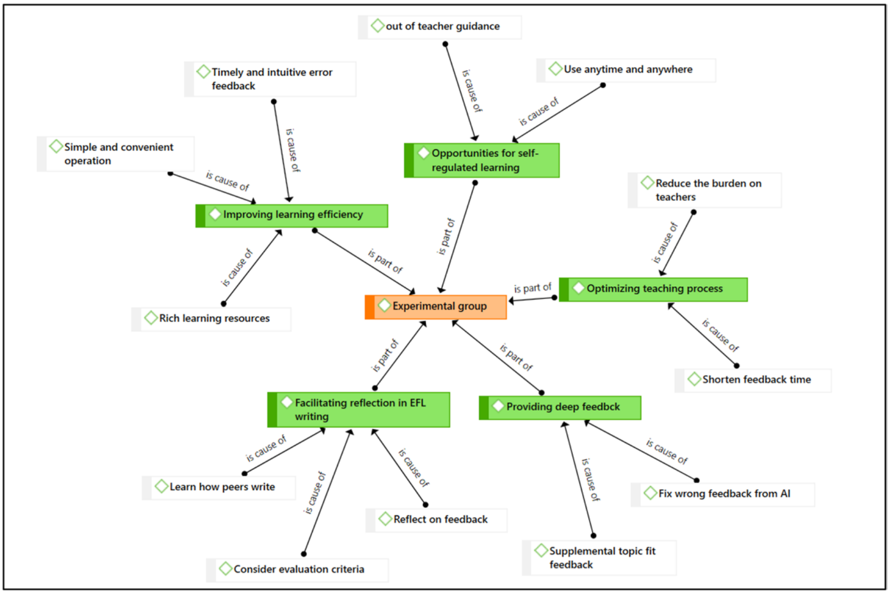

Fig. 8 A conceptual framework for learning gains of students in the experimental group

AWE tools would tell me more clearly what errors I had, and it would be quick." [E10-10:22–23] (This code, in all its instances, represents the student's number, the ID number of the coded content, and the location of the code). Another student reported: "The AWE gives feedback on the spot, where it could have taken days to get feedback from a teacher if they were looking for corrections, which greatly reduced the waiting time." [E03-3:14–17]. A student mentioned in the interview that "using the AWE tools to practice writing can reduce teacher workload." [C05-5:11–12].

5.5.2 Theme-2: Improving learning efficiency

The interactive interface of the AWE learning platform (Frequency of the experimental group: 3; Frequency of the control group: 1) and the rich learning resources (Frequency of the experimental group: 6; Frequency of the control group: 8) enabled students to receive timely and rapid feedback on their errors when writing with the AWE tools (Frequency of the experimental group: 9; Frequency of the control group: 7), which significantly improved their learning efficiency. Most of the students said, "The AWE not only provided me with feedback on vocabulary and grammatical errors, but also provided alternative expressions for sentences and reference examples that can help me make better revisions." [C10-10:7–14]. Other students said, "The operation on the AWE tools is convenient compared to writing on paper where you have to wait a long time for feedback from the teacher." [E05-5:5–7]. In addition, most learners also mentioned that "it was able to point out any errors in my writing in a timely and quick manner." [E-07–7:3–4].

5.5.3 Theme-3: Opportunities for self-regulated learning

Learners benefit from the fact that AWE tools can be used without teacher guidance (frequency of the experimental group: 3; frequency of the control group: 3) and it allows for more self-regulated learning opportunities for students at any time or place (Frequency of the experimental group: 3; Frequency of the control group: 3). Students mentioned in the interview that "After school, I could use it to practice my writing, and at the same time it would help me compose my essay better when there is no time limit." [E06-6:19–20]. Other students noted that "it was able to give me timely help and guidance when I wanted to know about my writing mistakes without the help of a teacher." [C11-11:13–16].

5.5.4 Theme-4: Facilitating reflection in EFL writing

Using the PA-AWE approach, the students in the experimental group were able to deeply think about the rubrics (Frequency of the experimental group: 4), learn how peers write (Frequency of the experimental group: 6), and reflect on feedback and revision suggestions from the machine and peers (Frequency of the experimental group: 6), which greatly facilitated deeper thinking about how to write. On the contrary, students in the control group only reflected on how to revise their writing by reflecting on the automatic feedback given by the machine (frequency of the control group: 3). One student in the experimental group mentioned in the interview that "When I assess my peer, I will think about his strengths and weaknesses from different perspectives and make up for them according to the teacher's rubrics." [E04-4:5–6]. Other students stated: "When I give comments to others, I reflect on whether I have the same problem." [E08-8:4–5]. In addition, students from both groups mentioned in the interview that "the feedback on vocabulary and grammar, as well as the alternative expressions, could help me optimize my writing and improve my writing skills." [C03-3:3–7].

5.5.5 Theme-5: Providing deep feedback

Considering that the AWE gives less feedback on topic matching and fails to identify errors and even provides inappropriate feedback sometimes, the PA has the potential to remedy the machine's incorrect feedback (Frequency of the experimental group: 3) and provide additional feedback on topic matching (Frequency of the experimental group: 4). Most of the students in the experimental group reported the positive impact of PA, for instance, one student said in the interview: "In some fixed-form letter writing, I made some mistakes which could not be recognized by the machine. However, my peers were able to help me identify the error at a glance." [E01-1:5–8]. Another student said, "My peer was more emotional than the computer and would give me feedback that fit my writing style and tell me if the connections between paragraphs were off the writing theme." [E06-6:8–10]. One student said in the interview, "Sometimes the AWE also made mistakes, and peers were able to help point them out very well." [E04-4:18–20].

6 Conclusion and discussion

The positive impact of AWE on EFL writing has been validated by previous studies; however, without the support of appropriate teaching strategies, the application of AWE in EFL instruction can still be limited. Therefore, this study proposed a PA-AWE approach to examine whether PA can be used as an effective pedagogical scaffold to facilitate learners' reflection in the writing process, and thus help students to make better use of AWE to improve their writing skills. The results revealed that the PA-AWE approach improved students' writing performance, learning motivation, and critical thinking, and significantly reduced their writing anxiety. Furthermore, the findings suggest that the PA-AWE approach not only helps students to receive timely and effective feedback on their writing, but also facilitates learners' reflection on the content of the automated feedback, thus improving the quality of revision and writing performance.

In response to research question 1, students in the PA-AWE group outperformed the C-AWE group in terms of writing performance. The possible reason for this is that PA helps learners observe peers' writing and to reflect on the rubrics. Learners are motivated to review their work carefully, thus improving the quality of their writing. This is consistent with previous research findings, which indicated that students could compare their learning performance with peers using the PA strategy (Schunn et al., 2016) and gain inspiration by appreciating or commenting on others to reflect on and improve their learning performance (Panadero & Alqassab, 2019; Seifert & Feliks, 2019). Furthermore, the latest findings of Tan et al. (2022) also suggested that comments from peers during the writing process help improve students' writing accuracy because, compared to the feedback provided by AWE, feedback from peers may be more effective in helping students write more sentences and produce more vocabulary items and word types (Ebadi & Rahimi, 2017; Matsumura & Hann, 2004). Furthermore, as Ma (2020) stated, positive feedback from peers can motivate and encourage students to work harder on revising their writing compared to AWE feedback. Moreover, when giving critical suggestions for peers' writing, peers not only focus on the linguistic aspects (Arnold et al., 2012), but also the content and relevance aspects (Ma, 2020), and reflect on their writing so that they are more successful in revising their work to achieve higher levels of writing performance. It can be seen that, as Shang (2022) and Tan et al. (2022) demonstrated, compared to the conventional single use of AWE to facilitate EFL writing, the integration of PA into AWE in this study was effective in helping learners to revise their work more carefully after receiving feedback from peers, and thus to have a deeper reflection on the content of the automatic feedback. Therefore, the PA-AWE approach proposed in this study provided new support and research perspectives to apply AWE in EFL writing instruction.

For research question 2, in terms of learning motivation, students in the PA-AWE group scored higher than those in the C-AWE group. This may be because students promote classroom engagement by completing PA tasks. In this process, the students consciously compared their work with that of their peers and received better quality comments and suggestions for revision, which motivated them to write. This echoes previous research in which peer feedback was considered an effective way to encourage interaction and collaboration (Fu et al., 2022), and through PA, learners were able to comment more on peers' work and to revise their own work more successfully, thereby promoting learner motivation and engagement (Shih, 2011). This finding seems to differ from the results of Cho and Cho (2011), who showed that the more comments received, the lower the quality of the final writing. However, the findings of Cheng et al. (2015) supported the findings of this study that positive comments from peers, such as expressing positive feelings about peers' work, would help to promote the development of better work. During the PA process, comments on the advantages of writing would motivate and encourage learners to revise their writing (Hattie & Timperley, 2007), thus promoting learning motivation (Shintani, 2016).

For research question 3, in terms of critical thinking, the PA-AWE has a higher tendency than C-AWE. Probably the reason for this is that the PA strategy somewhat prompted the learners to reflect deeply after receiving feedback from the AWE. When learners acted as assessors to give high-quality feedback and comments on their peers' work, they would think critically about the strengths and weaknesses of their peers' writing from different perspectives based on the rubrics given by the teacher. When they are in the role of assessee, they would reflectively adopt the automatic feedback based on their peers' suggestions. The study by Jiang et al. (2022) supported the argument that PA could prompt students to make thoughtful comments and provide feedback on peers' work based on previously defined standards and to interpret, analyze, and reflect upon the feedback received from peers. This is also supported by Jeong's (2003) findings that comments involving specific suggestions and in-depth interaction may lead to cognitive conflicts and stimulate mutual analysis, evaluation, induction, and revision among students, thus facilitating the development of critical thinking skills. Van et al. (2006) drew similar conclusions. The contradictions between learners' existing experiences and new understandings make them doubt their beliefs and try new ideas. In contrast, the cognitive conflicts generated by social interactions between learners effectively contribute to students' intellectual development and develop their critical thinking skills. The PA-AWE approach is effective in helping learners improve their writing quality and skills by generating cognitive conflicts through mutual criticism, thus allowing them to reflect and learn.

In response to research question 4, students in the PA-AWE group showed lower levels of EFL writing anxiety than the C-AWE group. This is probably because, by using AWE for writing, the students were able to identify their errors in vocabulary, grammar, and phrases. The enriched learning resources of AWE (e.g., the provision of alternative expressions for words and phrases, as well as excellent examples) helped to reduce the psychological stress caused by the lack of vocabulary and poor use of grammar in writing. Also, through PA, when learners did not understand the meaning of the feedback from AWE, peers could give explanations and notes. This is in line with Shang's (2022) findings that AWE would alert students to errors when they were writing, so that they could immediately revise them; if students still felt unsure about the feedback provided by AWE, they could seek help from their peers after completing the essay. Furthermore, AWE feedback provided learners with the opportunity for repeated practices, which effectively reduced learners' anxiety due to the fear of making mistakes during the writing process and improved writing skills (Cotos, 2015; Stevenson & Phakiti, 2014). The findings of Tan et al. (2022) also supported the result that the combined approach of PA and AWE could essentially help EFL learners to solve most of the problems encountered in writing. In this PA-AWE approach, learners were more inclined to focus on the global problems (e.g., content and meaning) after receiving feedback from AWE on language, which provided learners with more aspects of revision suggestions, making them feel comfortable and at ease on the writing platform.

Although the PA-AWE approach proposed in this study showed positive impacts on EFL writing instruction, there are still some limitations. First, since the experiment was conducted over a short period of 4 weeks and the development of writing habits and skills is a long-term cumulative process, future studies could attempt to extend the experimental duration. Second, the sample size of this study was small and individual factors such as gender and educational background were not considered. Third, it should be noted that it is a bit challenging for students who are not computer proficient at using AWE for EFL writing learning. Finally, the quality of PA has not been controlled in this study, which can result in some students failing to receive feedback from their peers in an effective and timely manner.

In summary, the PA-AWE approach proposed in this study has two main contributions to EFL writing instruction: first, the use of the PA-AWE approach opened up new pathways for EFL writing instruction when teachers are overwhelmed with work; second, the PA-AWE approach helped students reflect on how to revise their writing and improve their writing skills, rather than glancing at the AWE feedback without deep reflection. Thus, future research suggestions are as follows: the integration of PA with other AI technology in other disciplines could be considered in the future; more variables such as the type of assignment, the learners' computer experience, and their motivation and learning interest should also be taken into account in future research.

Authors' contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Chen-Chen Liu, Shi-Jie Liu and Naini Wang. Project administration were performed by Gwo-Jen Hwang, Yun-Fang Tu and Youmei Wang. Methodology and supervision were performed Gwo-Jen Hwang and Yun-Fang Tu. The first draft of the manuscript was written by Chen-Chen Liu and Shi-Jie Liu. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding This research received grants from the National Office for Educational Sciences Planning (NO. BCA210086).

Data availability The data and materials are available upon request to the corresponding author.

Code availability Not applicable

Declarations

Ethics approval The ethical requirements for research in this selected university were followed.

Consent to participate The participants all agreed to take part in this study.

Consent for publication The publication of this study has been approved by all authors.

Conflicts of interest/Competing interests There is no potential conflict of interest in this study.

References

Andrade, H., & Du, Y. (2005). Student perspectives on rubric-referenced assessment. *Practical Assessment, Research, and Evaluation, 10*(1), 3. https://doi.org/10.7275/g367-ye94

Arnold, N., Ducate, L., & Kost, C. (2012). Collaboration or cooperation? Analyzing group dynamics and revision processes in wikis. *Calico Journal, 29*(3), 431–448. https://doi.org/10.11139/cj.29.3.431-448

Bai, L., & Hu, G. (2017). In the face of fallible AWE feedback: How do students respond? *Educational Psychology, 37*(1), 67–81. https://doi.org/10.1080/01443410.2016.1223275

Baker, B. A., Homayounzadeh, M., & Arias, A. (2020). Development of a test taker-oriented rubric: Exploring its usefulness for test preparation and writing development. *Journal of Second Language Writing, 50*, 100771. https://doi.org/10.1016/j.jslw.2020.100771

Barrot, J. S., & Agdeppa, J. Y. (2021). Complexity, accuracy, and fluency as indices of college-level L2 writers' proficiency. *Assessing Writing, 47*, 100510. https://doi.org/10.1016/j.asw.2020.100510

Bitchener, J. (2012). A reflection on 'the language learning potential' of written. *Journal of Second Language Writing, 21*(4), 348–363. https://doi.org/10.1016/j.jslw.2012.09.006

Calvo, R. A., & Ellis, R. A. (2010). Students' conceptions of tutor and automated feedback in professional writing. *Journal of Engineering Education, 99*(4), 427–438. https://doi.org/10.1002/j.2168-9830.2010.tb01072.x

Challob, A. I. (2021). The effect of flipped learning on EFL students' writing performance, autonomy, and motivation. *Education and Information Technologies, 26*(4), 3743–3769.

Chang, C. F. (2012). Peer review via three modes in an EFL writing course. *Computers and Composition, 29*(1), 63–78. https://doi.org/10.1016/j.compcom.2012.01.001

Chang, S. C., Hsu, T. C., & Jong, M. S. Y. (2020). Integration of the peer assessment approach with a virtual reality design system for learning earth science. *Computers & Education, 146*, 103758. https://doi.org/10.1016/j.compedu.2019.103758

Chen, C. H. (2010). The implementation and evaluation of a mobile self-and peer-assessment system. *Computers & Education, 55*(1), 229–236. https://doi.org/10.1016/j.compedu.2010.01.008

Cheng, K. H., Liang, J. C., & Tsai, C. C. (2015). Examining the role of feedback messages in undergraduate students' writing performance during an online peer assessment activity. *The Internet and Higher Education, 25*, 78–84. https://doi.org/10.1016/j.iheduc.2015.02.001

Cheng, Y. S. (2017). Development and preliminary validation of four brief measures of L2 language-skill-specific anxiety. *System, 68*, 15–25. https://doi.org/10.1016/j.system.2017.06.009

Cho, K., Schunn, C. D., & Wilson, R. W. (2006). Validity and reliability of scaffolded peer assessment of writing from instructor and student perspectives. *Journal of Educational Psychology, 98*(4), 891. https://doi.org/10.1037/0022-0663.98.4.891

Cho, Y. H., & Cho, K. (2011). Peer reviewers learn from giving comments. *Instructional Science, 39*(5), 629–643. https://doi.org/10.1007/s11251-010-9146-1

Cohen, J. (1988). *Statistical power analysis for the social sciences* (2nd ed.). Lawrence Erlbaum Associates.

Cotos, E. (2015). AWE for writing pedagogy: From healthy tension to tangible prospects. Special issue on assessment for writing and pedagogy. *Writing & Pedagogy, 7*(2–3), 197–231. https://doi.org/10.1558/wap.v7i2-3.26381

Creswell, J. W., & Poth, C. N. (2016). Qualitative inquiry and research design: Choosing among five approaches. *Sage Publications*. https://doi.org/10.1177/1524839915580941

Diab, N. M. (2010). Effects of peer-versus self-editing on students' revision of language errors in revised drafts. *System, 38*(1), 85–95. https://doi.org/10.1016/j.system.2009.12.008

Dikli, S., & Bleyle, S. (2014). Automated essay scoring feedback for second language writers: How does it compare to instructor feedback? *Assessing Writing, 22*, 1–17. https://doi.org/10.1016/j.asw.2014.03.006

Ebadi, S., & Rahimi, M. (2017). Exploring the impact of online peer-editing using Google Docs on EFL learners' academic writing skills: A mixed methods study. *Computer Assisted Language Learning, 30*(8), 787–815. https://doi.org/10.1080/09588221.2017.1363056

Ferris, D. R. (2010). Second language writing research and written corrective feedback in SLA: Intersections and practical applications. *Studies in Second Language Acquisition, 32*(2), 181–201. https://doi.org/10.1017/S0272263109990490

Filippou, K., Kallo, J., & Mikkilä-Erdmann, M. (2021). Supervising master's theses in international master's degree programmes: Roles, responsibilities and models. *Teaching in Higher Education*. https://doi.org/10.1080/13562517.2019.1636220

Fu, Q. K., Zou, D., Xie, H., & Cheng, G. (2022). A review of AWE feedback: types, learning outcomes, and implications. *Computer Assisted Language Learning*, 1–43. https://doi.org/10.1080/09588221.2022.2033787

Grimes, D., & Warschauer, M. (2010). Utility in a fallible tool: A multi-site case study of automated writing evaluation. *The Journal of Technology*, *Learning and Assessment*, *8*(6). Retrieved from https://ejournals.bc.edu/index.php/jtla/article/view/1625

Guardado, M., & Shi, L. (2007). ESL students' experiences of online peer feedback. *Computers and Composition, 24*(4), 443–461. https://doi.org/10.1016/j.compcom.2007.03.002

Guo, Q., Feng, R., & Hua, Y. (2021). How effectively can EFL students use automated written corrective feedback (AWCF) in research writing?. *Computer Assisted Language Learning*, 1–20. https://doi.org/10.1080/09588221.2021.1879161

Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research, 77*(1), 81–112. https://doi.org/10.3102/003465430298487

Ho, M. C., & Savignon, S. J. (2007). Face-to-face and computer-mediated peer review in EFL writing. *CALICO Journal*, 269–290. https://doi.org/10.1558/cj.v24i2.269-290

Hwang, G. J., Yang, T. C., Tsai, C. C., & Yang, S. J. H. (2009). A context-aware ubiquitous learning environment for conducting complex science experiments. *Computers & Education, 53*(2), 402–413. https://doi.org/10.1016/j.compedu.2009.02.016

Hyland, K., & Hyland, F. (2006). Feedback on second language students writing. *Language Teaching, 39*(2), 83–101. https://doi.org/10.1017/S0261444806003399

Jeong, A. C. (2003). The sequential analysis of group interaction and critical thinking in online. *The American Journal of Distance Education, 17*(1), 25–43. https://doi.org/10.1207/S15389286AJDE1701_3

Jiang, J. P., Hu, J. Y., Zhang, Y. B., & Yin, X. C. (2022). Fostering college students' critical thinking skills through peer assessment in the knowledge building community. *Interactive Learning Environments*, 1–17. https://doi.org/10.1080/10494820.2022.2039949

Jin, T., Jiang, Y., Gu, M. M., & Chen, J. (2022). "Their encouragement makes me feel more confident": Exploring peer effects on learner engagement in collaborative reading of academic texts. *Journal of English for Academic Purposes*, 101177. https://doi.org/10.1016/j.jeap.2022.101177

Jones, R. H., Garralda, A., Li, D. C., & Lock, G. (2006). Interactional dynamics in on-line and face-to-face peer-tutoring sessions for second language writers. *Journal of Second Language Writing, 15*(1), 1–23. https://doi.org/10.1016/j.jslw.2005.12.001

Lai, C. L., & Hwang, G. J. (2015). An interactive peer-assessment criteria development approach to improving students' art design performance using handheld devices. *Computers & Education, 85*, 149–159. https://doi.org/10.1016/j.compedu.2015.02.011

Latifi, S., & Noroozi, O. (2021). Supporting argumentative essay writing through an online supported peer-review script. *Innovations in Education and Teaching International, 58*(5), 501–511. https://doi.org/10.1080/14703297.2021.1961097

Latifi, S., Noroozi, O., Hatami, J., & Biemans, H. J. (2021). How does online peer feedback improve argumentative essay writing and learning? *Innovations in Education and Teaching International, 58*(2), 195–206. https://doi.org/10.1080/14703297.2019.1687005

Li, H., Xiong, Y., Hunter, C. V., Guo, X., & Tywoniw, R. (2020). Does peer assessment promote student learning? A meta-analysis. *Assessment & Evaluation in Higher Education, 45*(2), 193–211. https://doi.org/10.1080/02602938.2019.1620679

Li, J., Huang, J., & Cheng, S. (2022). The reliability, effectiveness, and benefits of peer assessment in college EFL speaking classrooms: Student and teacher perspectives. *Studies in Educational Evaluation, 72*, 101120. https://doi.org/10.1016/j.stueduc.2021.101120

Li, J., Link, S., & Hegelheimer, V. (2015). Rethinking the role of automated writing evaluation (AWE) feedback in ESL writing instruction. *Journal of Second Language Writing, 27*, 1–18. https://doi.org/10.1016/j.jslw.2014.10.004

Li, L., Liu, X., & Steckelberg, A. L. (2010). Assessor or assessee: How student learning improves by giving and receiving peer feedback. *British Journal of Educational Technology, 41*(3), 525–536. https://doi.org/10.1111/j.1467-8535.2009.00968.x

Liao, H. C. (2016). Using automated writing evaluation to reduce grammar errors in writing. *Elt Journal, 70*(3), 308–319. https://doi.org/10.1093/elt/ccv058

Lin, H. C., Hwang, G. J., Chang, S. C., & Hsu, Y. D. (2021). Facilitating critical thinking in decision making-based professional training: An online interactive peer-review approach in a flipped learning context. *Computers & Education, 173*, 104266. https://doi.org/10.1016/j.compedu.2021.104266

Lin, Y. N., Hsia, L. H., Sung, M. Y., & Hwang, G. H. (2019). Effects of integrating mobile technology-assisted peer assessment into flipped learning on students' dance skills and self-efficacy. *Interactive Learning Environments, 27*(8), 995–1010. https://doi.org/10.1080/10494820.2018.1461115

Link, S., Mehrzad, M., & Rahimi, M. (2022). Impact of automated writing evaluation on teacher feedback, student revision, and writing improvement. *Computer Assisted Language Learning, 35*(4), 605–634. https://doi.org/10.1080/09588221.2020.1743323

Liou, H. C., & Peng, Z. Y. (2009). Training effects on computer-mediated peer review. *System, 37*(3), 514–525. https://doi.org/10.1016/j.system.2009.01.005

Liu, C., Hou, J., Tu, Y. F., Wang, Y., & Hwang, G. J. (2021). Incorporating a reflective thinking promoting mechanism into artificial intelligence-supported English writing environments. *Interactive Learning Environments*, 1–19. https://doi.org/10.1080/10494820.2021.2012812

Ma, Q. (2020). Examining the role of inter-group peer online feedback on wiki writing in an EAP context. *Computer Assisted Language Learning, 33*(3), 197–216. https://doi.org/10.1080/09588221.2018.1556703

Matsumura, S., & Hann, G. (2004). Computer anxiety and students' preferred feedback methods in EFL writing. *The Modern Language Journal, 88*(3), 403–415. https://doi.org/10.1111/j.0026-7902.2004.00237.x

Noroozi, O., & Hatami, J. (2018). The effects of online peer feedback and epistemic beliefs on students' argumentation-based learning. *Innovations in Education and Teaching International*. https://doi.org/10.1080/14703297.2018.1431143

Noroozi, O., Hatami, J., Bayat, A., van Ginkel, S., Biemans, H. J., & Mulder, M. (2020). Students' online argumentative peer feedback, essay writing, and content learning: Does gender matter? *Interactive Learning Environments, 28*(6), 698–712. https://doi.org/10.1080/10494820.2018.1543200

Panadero, E., & Alqassab, M. (2019). An empirical review of anonymity effects in peer assessment, peer feedback, peer review, peer evaluation and peer grading. *Assessment & Evaluation in Higher Education*. https://doi.org/10.1080/02602938.2019.1600186

Pham, H. T. P. (2020). Computer-mediated and face-to-face peer feedback: student feedback and revision in EFL writing. *Computer Assisted Language Learning*, 1–37. https://doi.org/10.1080/09588221.2020.1868530

Ranalli, J. (2018). Automated written corrective feedback: How well can students make use of it? *Computer Assisted Language Learning, 31*(7), 653–674. https://doi.org/10.1080/09588221.2018.1428994

Ranalli, J., Link, S., & Chukharev-Hudilainen, E. (2017). Automated writing evaluation for formative assessment of second language writing: Investigating the accuracy and usefulness of feedback as part of argument-based validation. *Educational Psychology, 37*(1), 8–25. https://doi.org/10.1080/01443410.2015.1136407

Scardamalia, M., & Bereiter, C. (1994). Computer support for knowledge-building communities. *The Journal of the Learning Sciences, 3*(3), 265–283. https://doi.org/10.1207/s15327809jls0303_3

Schunn, C., Godley, A., & DeMartino, S. (2016). The reliability and validity of peer review of writing in high school AP English classes. *Journal of Adolescent & Adult Literacy, 60*(1), 13–23. https://doi.org/10.1002/jaal.525

Seifert, T., & Feliks, O. (2019). Online self-assessment and peer-assessment as a tool to enhance student-teachers' assessment skills. *Assessment & Evaluation in Higher Education, 44*(2), 169–185. https://doi.org/10.1080/02602938.2018.1487023

Shang, H. F. (2022). Exploring online peer feedback and automated corrective feedback on EFL writing performance. *Interactive Learning Environments, 30*(1), 4–16. https://doi.org/10.1080/10494820.2019.1629601

Shih, R. C. (2011). Can Web 2.0 technology assist college students in learning English writing? Integrating Facebook and peer assessment with blended learning. *Australasian Journal of Educational Technology*, *27*(5). https://doi.org/10.14742/ajet.934

Shintani, N. (2016). The effects of computer-mediated synchronous and asynchronous direct corrective feedback on writing: A case study. *Computer Assisted Language Learning, 29*(3), 517–538. https://doi.org/10.1080/09588221.2014.993400

Stevenson, M., & Phakiti, A. (2014). The effects of computer-generated feedback on the quality of writing. *Assessing Writing, 19*, 51–65. https://doi.org/10.1016/j.asw.2013.11.007

Tan, S., Cho, Y. W., & Xu, W. (2022). Exploring the effects of automated written corrective feedback, computer-mediated peer feedback and their combination mode on EFL learner's writing performance. *Interactive Learning Environments*, 1–11. https://doi.org/10.1080/10494820.2022.2066137

Topping, K. (1998). Peer assessment between students in colleges and universities. *Review of Educational Research, 68*(3), 249–276. https://doi.org/10.3102/00346543068003249

Tsivitanidou, O. E., Zacharia, Z. C., & Hovardas, T. (2011). Investigating secondary school students' unmediated peer assessment skills. *Learning and Instruction, 21*(4), 506–519. https://doi.org/10.1016/j.learninstruc.2010.08.002

Tuzi, F. (2004). The impact of e-feedback on the revisions of L2 writers in an academic writing course. *Computers and Composition, 21*(2), 217–235. https://doi.org/10.1016/j.compcom.2004.02.003

Van Beuningen, C. G., De Jong, N. H., & Kuiken, F. (2012). Evidence on the effectiveness of comprehensive error correction in second language writing. *Language Learning, 62*(1), 1–41. https://doi.org/10.1111/j.1467-9922.2011.00674.x

Wang, L. C., & Chen, M. P. (2010). The effects of game strategy and preference-matching on flow experience and programming performance in game-based learning. *Innovations in Education and Teaching International, 47*(1), 39–52. https://doi.org/10.1080/14703290903525838

Wang, Y. J., Shang, H. F., & Briody, P. (2013). Exploring the impact of using automated writing evaluation in English as a foreign language university students' writing. *Computer Assisted Language Learning, 26*(3), 234–257. https://doi.org/10.1080/09588221.2012.655300

Warschauer, M., & Grimes, D. (2008). Automated writing assessment in the classroom. *Pedagogies: An International Journal, 3*(1), 22–36. https://doi.org/10.1080/15544800701771580

Warschauer, M., & Ware, P. (2006). Automated writing evaluation: Defining the classroom research agenda. *Language Teaching Research, 10*(2), 157–180. https://doi.org/10.1191/1362168806lr190oa

Wu, W. C. V., Petit, E., & Chen, C. H. (2015). EFL writing revision with blind expert and peer review using a CMC open forum. *Computer Assisted Language Learning, 28*(1), 58–80. https://doi.org/10.1080/09588221.2014.937442

Yang, Y. F. (2011). A reciprocal peer review system to support college students' writing. *British Journal of Educational Technology, 42*(4), 687–700. https://doi.org/10.1111/j.1467-8535.2010.01059.x

Zhai, N., & Ma, X. (2021). Automated writing evaluation (AWE) feedback: A systematic investigation of college students' acceptance. *Computer Assisted Language Learning*, 1–26. https://doi.org/10.1080/09588221.2021.1897019

Zhang, Z. V., & Hyland, K. (2018). Student engagement with teacher and automated feedback on L2 writing. *Assessing Writing, 36*, 90–102. https://doi.org/10.1016/j.asw.2018.02.004

Authors and Affiliations

Chen-Chen Liu[1] · Shi-Jie Liu[1] · Gwo-Jen Hwang[2,3] · Yun-Fang Tu[4] · Youmei Wang[1] · Naini Wang[5]

Chen-Chen Liu
lcc5424548@126.com

Shi-Jie Liu
liusj223@gmail.com

Yun-Fang Tu
sandy0692@gmail.com

Youmei Wang
wangyoumei@126.com

Naini Wang
nany98@126.com

[1] Department of Educational Technology, Wenzhou University, Wenzhou, People's Republic of China

[2] Graduate Institute of Digital Learning and Education, National Taiwan University of Science and Technology, Taipei, Taiwan

[3] Graduate Institute of Educational Information and Measurement, National Taichung University of Education, Taichung, Taiwan

[4] Department of Library and Information Science, Research and Development Center for Physical Edu-Cation, Health, and Information Technology, Fu Jen Catholic University, New Taipei City, Taiwan

[5] School of Foreign Languages, Wenzhou University of Technology, Wenzhou, People's Republic of China